Software Engineering Department
Braude College

Capstone Project Phase A – 61998

WorkWithHelmtOnly

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Abstract

Ensuring workplace safety requires innovative solutions for monitoring workers' compliance with safety standards. This project focuses on developing an application capable of detecting the safety status of workers in photos and videos. Specifically, the system identifies workers and evaluates their use of essential safety tools, such as helmets and reflective jackets. Using YOLO and OpenCV, the app processes both still images and video frames. The primary objective is to create an accurate and practical tool for workplace safety monitoring, capable of enhancing compliance and reducing risks.

Keywords: Workplace safety monitoring, Computer vision, YOLO

1. Introduction

Ensuring workplace safety is a critical priority, particularly in high-risk environments such as construction sites and industrial settings. Compliance with safety protocols, such as wearing helmets and vests, significantly reduces the risk of injuries and promotes a secure working environment. However, monitoring adherence to these protocols across large or remote workplaces can be challenging, often requiring innovative approaches to ensure safety standards are met.

Deep learning has revolutionized computer vision by enabling machines to learn patterns directly from data, with convolutional neural networks (**CNNs**) at its core. CNNs process image data through convolutional layers to extract features like edges and textures, while pooling layers reduce dimensionality to enhance efficiency. In this project, CNNs are used to detect helmets and vests by training on labeled datasets, enabling accurate differentiation between compliant and non-compliant workers, ensuring effective workplace safety monitoring.

Our project addresses these challenges by introducing an application designed to analyze video footage from workplace cameras. The application processes captured video to detect workers, identify their compliance with safety gear requirements, and flag potential safety violations. The analyzed results, including instances of non-compliance, are stored in the system, allowing managers to review detailed reports and make informed decisions to enhance workplace safety. By automating the analysis of safety practices, our application reduces reliance on manual inspections, minimizes human error, and provides actionable insights. Through the application of advanced technologies such as computer vision and machine learning, this project offers a scalable and cost-effective solution to workplace safety challenges. It not only aids in maintaining compliance but also contributes to a culture of accountability and continuous improvement in high-risk industries.

Additionally, a prototype of the system has been developed to demonstrate its functionalities, including user-friendly interfaces such as the dashboard for compliance monitoring, video review for detecting safety violations, and pages for generating reports and managing settings. These prototypes showcase the system's practical application and its ability to facilitate workplace safety monitoring effectively.

2. Background And Related Work

The increasing integration of technology into workplace environments has paved the way for innovative safety monitoring applications. These applications include systems that leverage advancements in machine learning and computer vision to ensure worker safety. For example, research on worker video analysis has explored motion amount quantification, which contributes to better monitoring and understanding of workplace conditions [11]. Such technologies allow organizations to address safety challenges effectively by monitoring compliance with safety protocols in real-time or retrospectively.

These solutions often leverage advancements in machine learning and computer vision to enhance the detection and prevention of safety violations. For instance, convolutional neural networks (CNNs), which are foundational in computer vision, enable systems to process video streams and recognize objects such as helmets and vests [7]. Violations, such as the absence of required safety gear, are flagged for further action.

Numerous studies and applications have demonstrated the potential of such technologies to address compliance challenges in high-risk environments like construction and industrial sites. For instance, object detection models like YOLO have been widely adopted to detect PPE, enabling real-time monitoring of safety compliance [1]. These applications significantly reduce the need for manual inspections while improving efficiency and accuracy.

Researchers have utilized deep learning models, such as convolutional neural networks (CNNs), to detect helmets, vests, and other safety gear in images or videos. CNNs automatically extract and classify features, making them a critical component of object detection systems [7]. For instance, Redmon and Farhadi introduced YOLO, a real-time object detection system that has become a cornerstone of safety monitoring technology [1].

A study published in IEEE Access developed a model capable of identifying the presence of helmets and vests in real-time video streams, achieving high accuracy rates in controlled environments [5]. This approach highlights the growing trend of automating workplace safety inspections through image and video analysis.

Another area of advancement involves applications designed to assess safety compliance retrospectively. These solutions analyze recorded footage to identify safety violations and provide actionable insights. For instance, retrospective analysis frameworks have been developed to process archived videos, detect workers, and classify their activities, determining whether safety protocols were followed during specific tasks [10]. Such analyses have proven invaluable for identifying trends in non-compliance and informing safety training programs.

In industrial settings, similar applications have been deployed to evaluate worker behavior and adherence to safety guidelines. By combining object detection techniques with human pose estimation, these systems can assess whether workers maintain safe postures and avoid hazardous areas. Guidelines from organizations like OSHA emphasize the importance of consistently using PPE and adhering to safety protocols [12].

Studies have shown that these methods can not only detect unsafe conditions but also predict potential risks based on historical data, enabling proactive safety measures [5]. These advancements underscore the potential of leveraging video analysis for workplace safety. However, challenges remain, including the need for diverse datasets to train robust models, addressing privacy concerns in video monitoring, and ensuring scalability for large or remote workplaces. The proposed application builds upon these existing works, focusing on retrospective video analysis to provide managers with detailed reports and actionable insights, further enhancing safety compliance in high-risk industries.

2.1. Computer Vision and OpenCV

Computer vision, a critical field within artificial intelligence, allows machines to interpret and understand visual data, bridging the gap between human perception and computational analysis. It provides the foundation for technologies like object detection and real-time video processing, which are essential in safety monitoring systems. Among the tools available for implementing computer vision, OpenCV stands out as one of the most versatile and widely used libraries [13]. First introduced by Intel in 2000, OpenCV has become a standard in the field, offering powerful tools for real-time image and video analysis. Its applications span diverse domains, from facial recognition and autonomous vehicles to industrial safety monitoring [3].

In the context of workplace safety, OpenCV proves to be a game-changer. Its ability to process live video streams efficiently makes it particularly suitable for detecting critical safety equipment such as helmets and vests in dynamic and potentially hazardous environments. For example, OpenCV's robust features ensure accurate detection of safety violations, such as the absence of helmets, by integrating seamlessly with deep learning models like YOLO [1]. By employing image preprocessing techniques such as noise reduction and edge detection, OpenCV optimizes video frames for clearer analysis [14].

2.2. Object detection (OD)

Object detection is a computer vision technique that identifies and locates objects within images or videos, assigning them specific labels and bounding boxes. This process enables systems to recognize multiple objects simultaneously and determine their positions. Modern object detection methods predominantly utilize deep learning, especially Convolutional Neural Networks (CNNs), to automatically learn hierarchical features from raw images, enhancing detection accuracy [7] (Fig. 1). Notable deep learning-based object detection frameworks include R-CNN [8], YOLO [1], and SSD [9].

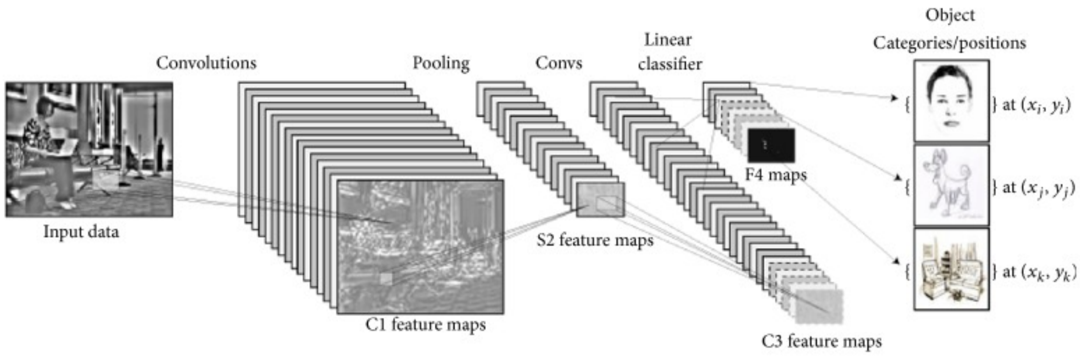

Figure 1. Workflow of a Convolutional Neural Network (CNN) for Object Detection. This figure illustrates the typical workflow of a Convolutional Neural Network (CNN) for object detection. The process begins with input data, which undergoes feature extraction through convolutional layers and pooling layers. These features are then classified using fully connected layers and linear classifiers to produce object detection outputs, including object positions and categories. In this project, the approach is adapted to detect safety equipment, such as helmets and vests, ensuring accurate compliance monitoring in workplace environments.

2.3. Object Detection With YOLO

YOLO is a real-time object detection system that uses a single-stage detection approach. YOLO, introduced in 2016 by Redmon and Farhadi, is known for its speed, efficiency, and high accuracy compared to other object detection methods [1]. It processes images in a single pass, identifying objects and drawing rectangular bounding boxes with corresponding labels. YOLO performs exceptionally well on the COCO dataset, which contains a wide variety of annotated images and videos [3]. Its ability to detect fast-moving objects quickly makes it ideal for real-time applications, focusing on both speed and performance [4]. The general scheme of YOLO is illustrated in **Fig. 2**.

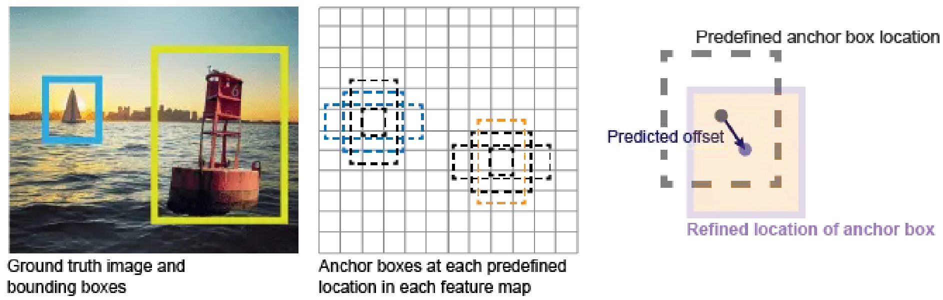

Figure 2. Visualization of YOLO Object Detection Algorithm Workflow. The YOLO object detection algorithm divides an image into a grid and predicts bounding boxes and associated class probabilities for each cell. The image on the left illustrates detected objects with their bounding boxes, while the center grid represents the division of the image for object localization. The rightmost section demonstrates how anchor boxes are refined and adjusted to align with predicted object locations. This efficient grid-based approach allows YOLO to achieve real-time object detection, which is a key component of our project for identifying safety equipment like helmets and vests.

2.4. Convolutional Neural Networks (CNNs)

Convolutional Neural Networks (CNNs) are a class of deep learning models designed to process data with a grid structure, like images. Inspired by the human visual system, they excel at extracting spatial features and patterns. A CNN has key components: convolutional layers apply filters to detect features like edges and textures, activation functions (e.g., ReLU) add non-linearity, and pooling layers reduce the size of feature maps to make the model more efficient and robust. Finally, fully connected layers aggregate the extracted features for tasks like classification or regression. CNNs automate feature extraction, making them ideal for computer vision tasks like image recognition, object detection, and facial recognition.

Architecture of a convolutional neural network (CNN) involves several sequential steps (Fig. 3): The process starts with the input image, which passes through convolutional layers where features such as edges and patterns are extracted using kernels. Max-pooling layers follow, reducing the spatial dimensions while preserving essential features. The feature maps are then flattened and passed through fully connected layers with activation functions to generate output predictions.

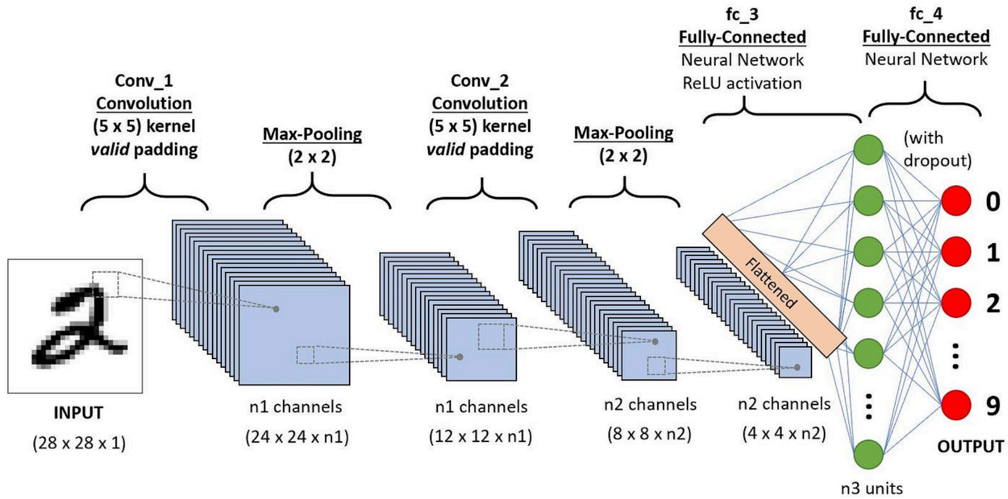

Figure 3. Architecture of a Convolutional Neural Network (CNN) Workflow

2.5 Non-Maximum Suppression (NMS)

NMS is a technique used in object detection to eliminate redundant overlapping bounding boxes, retaining only the most relevant ones. By suppressing non-maximum boxes based on confidence scores and overlap thresholds, NMS refines detection results, ensuring that each object is accurately localized without multiple overlapping boxes (Fig. 4).

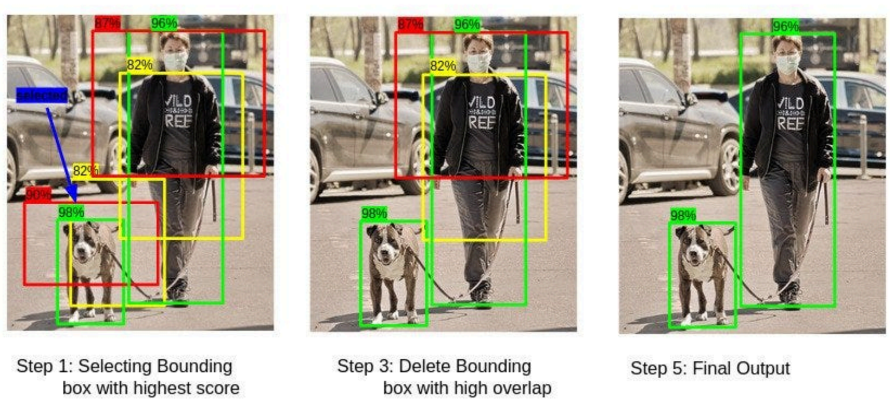

Figure 4. Illustration of Non-Maximum Suppression (NMS) in Object Detection". Non-Maximum Suppression (NMS) is a critical post-processing step in object detection that eliminates redundant bounding boxes. In the first step, the algorithm selects the bounding box with the highest confidence score. It then compares this box with others that have high overlap (based on Intersection over Union, IoU), suppressing those with lower confidence. The final output retains only the most accurate and relevant bounding boxes for each detected object.

2.6. Region-Based Convolutional Neural Networks (R-CNN)

R-CNN was one of the first object detection models to combine region proposals and deep learning [2]. The process starts with generating potential object regions using a selective search algorithm. These regions, around 2,000 in total, are resized and passed through a CNN to extract features. Then, classifiers like Support Vector Machines (SVMs) determine the category of objects, while a bounding box regressor fine-tunes the location of these objects. While R-CNN improved detection accuracy significantly, its method of processing each region individually made it very slow. This inefficiency paved the way for faster models like Fast R-CNN and Faster R-CNN (Fig. 5).

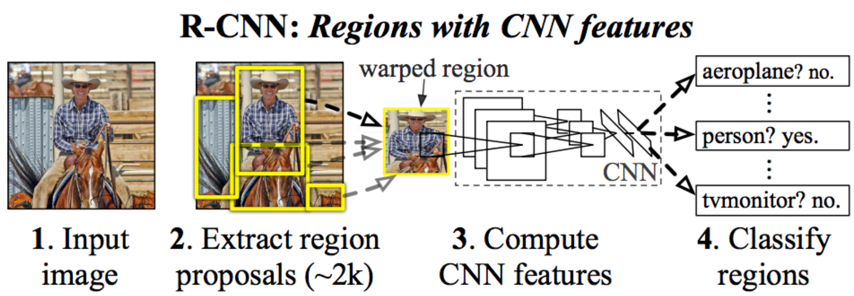

Figure 5. R-CNN Workflow for Object Detection.

2.8 Norfair

is a versatile Python library designed for object tracking in videos and images. It employs advanced algorithms to efficiently track objects across frames by associating detections over time. Norfair is highly adaptable and allows integration with various detection methods, making it suitable for applications such as monitoring safety compliance in workplaces, where detecting helmets, vests, or other safety tools is crucial. Its simplicity and modularity make it an ideal choice for projects requiring robust tracking capabilities [6].

Norfair works by estimating the future position of each point based on its past positions. It then tries to match these estimated positions with newly detected points provided by the detector. For this matching to occur, Norfair can rely on any distance function. There are some predefined distances already integrated in Norfair, and the users can also define their own custom distances. Therefore, each object tracker can be made as simple or as complex as needed. As an example we use Detectron2 to get the single point detections to use with this distance function. We just use the centroids of the bounding boxes it produces around cars as our detections, and get the following results.

Figure 6. Norfair tracks cars

3. Model

The workplace safety monitoring application is designed to detect and evaluate worker compliance with safety standards using a combination of YOLO for object detection, OpenCV for post-processing, and Norfair for object tracking in videos. This section details the key components of the model and their integration.

3.1 Object Detection with YOLO

The YOLO (You Only Look Once) model is the foundation for detecting workers and their safety equipment in both images and videos. YOLO's speed and accuracy make it ideal for real-time and batch processing tasks.
Key steps include:

- **Dataset Preparation:** A curated and labeled dataset containing images of workers wearing or missing safety equipment (helmets, reflective jackets, etc.) is used for training. Data augmentation techniques such as rotations, cropping, and varying lighting conditions improve model generalization.
- **Model Training:** YOLO is fine-tuned on the PPE dataset to optimize its ability to detect safety tools and differentiate between compliant and non-compliant states.
- **Inference:** YOLO outputs bounding boxes with class labels (e.g., "helmet" or "reflective jacket"), enabling the system to analyze compliance.

3.2 Post-Processing with OpenCV

OpenCV enhances the output from YOLO by performing additional tasks critical to the system's functionality:

- **Bounding Box Visualization:** Safety compliance is visualized using color-coded bounding boxes (green for compliance, red for missing equipment).
- **Preprocessing:** Image adjustments, such as resizing, denoising, and normalization, are applied to improve detection accuracy under suboptimal conditions.
- **Labeling Missing Equipment:** Detected non-compliance is explicitly labeled (e.g., "Helmet Missing") for clarity in reports and alerts.

3.3 Object Tracking with Norfair

Norfair is integrated for tracking detected objects across video frames. This ensures continuity in monitoring worker compliance, even in dynamic environments.
Norfair's role includes:

- **Worker Tracking:** Maintains consistent identities for workers across frames, even as they move or partially occlude each other.
- **Compliance History:** Tracks changes in safety compliance, such as a worker removing or putting on safety equipment during the video.
- **Event Logging:** Logs specific timestamps for non-compliance events, enabling detailed reporting and efficient resolution.

3.4 Compliance Evaluation

The final component evaluates the detected and tracked objects against predefined safety criteria. The system ensures:

- **Accurate Detection:** Each worker is assessed for the presence of mandatory safety tools.
- **Detailed Reports:** A comprehensive report is generated, summarizing compliance and highlighting non-compliance with timestamped details for video analysis.
- **Notifications:** Non-compliance triggers alerts, such as email notifications, with visual evidence (images or video snippets) attached.

3.5 Model Integration

All components—YOLO for detection, OpenCV for preprocessing and visualization, and Norfair for tracking—are seamlessly integrated into a Python-based application. The system is optimized for batch processing of images and real-time or near-real-time video analysis. Its modular design ensures scalability and adaptability for future enhancements, such as adding new safety tools or integrating real-time notifications (Fig 7):

1. The application starts, and the supervisor uploads a video.
2. The system processes the video, analyzing each frame to detect workers and check if they are wearing the required safety equipment, such as helmets and reflective jackets.
3. If all workers are compliant, nothing happens, and the process ends.
4. If a worker is detected without the necessary equipment, the system sends a notification to the supervisor with an image of the detected worker.
5. The supervisor reviews the image and decides whether the detection is correct.
6. If the detection is incorrect, the supervisor can refine the model by providing feedback, updating the dataset, and retraining the system to improve accuracy.
7. If the detection is correct, the supervisor can take action by instructing the worker to wear the required safety equipment.
8. The process continues for new videos, ensuring workplace safety through ongoing monitoring.

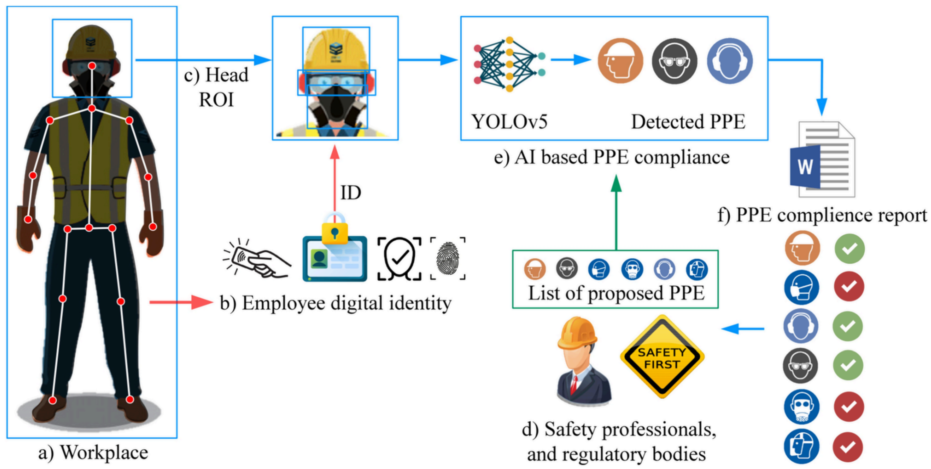

Figure 7. Integrated Workflow of AI-Based PPE Compliance Monitoring in Workplace Safety.

4. Dataset and training

4.1 Microsoft COCO Dataset

The "Microsoft COCO Dataset" is a large-scale dataset for object detection, segmentation, and captioning. It includes diverse images with multiple objects per scene. This dataset is essential for training AI models in computer vision.

4.2 PPE Dataset for Workplace Safety Computer Vision Project

The "PPE Dataset" is a robust and diverse collection of images designed for the development and enhancement of machine learning models in the realm of workplace safety. This dataset focuses on the detection and classification of various types of personal protective equipment (PPE) typically used in industrial and construction environments. The goal is to facilitate automated monitoring systems that ensure adherence to safety protocols, thereby contributing to the prevention of workplace accidents and injuries. Total Images: 1,613 images annotated for PPE detection. (https://universe.roboflow.com/siabar/ppe-dataset-for-workplace-safety/dataset/2)

Classes : BootsEar-protection, Glass, Glove, Helmet, Mask, Person, Vest.

Dataset Split:

Train Set: 70% (1,126 images)

Validation Set: 20% (326 images)

Test Set: 10% (161 images)

5. The Process

5.1 Research – Workplace Safety Monitoring

To build a reliable workplace safety monitoring system, we focused our research on the following questions:

- What are the most common safety violations in construction and industrial sites?
- How can object detection models improve safety compliance monitoring?
- What datasets are available for training safety gear detection models?
- What technologies and algorithms best suit real-time and retrospective safety monitoring?
- What challenges are associated with automated video analysis in workplace environments?

To address these questions, we reviewed scientific articles, industry guidelines, and relevant datasets such as the PPE Dataset for Workplace Safety [6]. Our research emphasized the importance of using advanced deep learning models and computer vision tools to ensure high accuracy in detecting compliance.

5.1.1 Constraints and Challenges – Workplace Safety Monitoring

Several constraints and challenges were identified during the research phase:

- **Environmental Variations:** Workplace lighting conditions, worker movement, and camera angles affect detection accuracy.
- **Data Availability:** Acquiring diverse and labeled datasets specific to safety gear detection was challenging.
- **Real-Time Processing:** Balancing detection speed and accuracy for real-time monitoring.
- **Privacy Concerns:** Managing video data while adhering to privacy regulations.
- **Hardware Limitations:** Ensuring compatibility with standard CCTV systems without significant hardware upgrades.

5.1.2 Conclusions from Research – Technology Selection

Based on our research, the following technologies and tools were chosen for the project:

- **YOLO (You Only Look Once)** for real-time object detection due to its speed and accuracy.
- **OpenCV** for image preprocessing and visualization of detected objects.
- **Norfair** for object tracking across multiple video frames.
- **Python and TensorFlow** for model training and inference.

These tools allow for efficient detection, tracking, and reporting of safety compliance.

5.2 Research – Object Detection Techniques

Object detection plays a vital role in our system as it enables the identification of workers and their protective equipment. The methods explored during our research include:

- **YOLO:** Real-time object detection with a single-stage architecture optimized for speed and performance.
- **SSD (Single Shot Multibox Detector):** Balances speed and accuracy for object detection tasks.
- **Faster R-CNN:** Region-based detection known for high accuracy but slower performance.

After comparison, YOLO was selected due to its balance of speed and accuracy, making it suitable for both real-time and retrospective analysis in safety monitoring.

5.3 Research – Tracking Methods

To ensure accurate worker tracking across frames, we explored various object tracking approaches, including:

- **Norfair:** A lightweight library designed for multi-object tracking, chosen for its simplicity and accuracy.
- **Kalman Filter:** Effective for motion prediction but less suitable for handling occlusions in dynamic environments.

Norfair was selected due to its integration with Python and suitability for the workplace safety use case.

5.4 Research – Compliance Evaluation Techniques

Compliance evaluation involves comparing detected safety equipment with predefined safety standards. Key techniques considered:

- **Rule-Based Evaluation:** Manually defining conditions for compliance, such as detecting both a helmet and vest.
- **ML-Based Evaluation:** Using a trained model to classify compliance states based on the detected objects.

Our system adopts a hybrid approach, using rule-based evaluations for simplicity and ensuring accuracy through the YOLO model.

5.5 Development Process

We adopted an **Agile development methodology**, focusing on iterative improvements and continuous feedback. The development phases included:

1. **Model Selection and Data Preparation:** Chose YOLO and prepared the PPE dataset for safety equipment detection.
2. **Object Detection Implementation:** Implemented YOLO for real-time detection of helmets and vests.
3. **Object Tracking Integration:** Incorporated Norfair for continuous tracking of workers across frames.
4. **Compliance Evaluation Logic:** Implemented a rule-based evaluation system for safety compliance assessment.
5. **User Interface Development:** Created a simple web-based dashboard for displaying compliance reports.
6. **Testing and Optimization:** Conducted multiple rounds of testing on various workplace scenarios and lighting conditions.

5.6 Testing and Evaluation Plan

To ensure the system performs as expected, we defined the following test cases:

- **Detection Accuracy:** Verify YOLO's ability to detect helmets and vests in varied conditions.
- **Tracking Consistency:** Ensure Norfair can track multiple workers without identity loss.
- **Compliance Reporting:** Confirm the system generates clear and actionable compliance reports.
- **Real-Time Performance:** Validate the system's ability to analyze live video feeds without significant delay.

6 Product

6.1 Requirements

Functional:

ID	Requirement	Description	Priority
FR1	Video Input Integration	The system must capture live video streams or accept uploaded recorded footage.	High
FR2	Preprocessing Video Frames	Resize, denoise, and normalize video frames for optimal detection accuracy.	High
FR3	Object Detection	Use YOLO to detect workers and safety equipment in images and video.	High
FR4	Safety Compliance Evaluation	Identify compliance and non-compliance based on detected safety equipment.	High
FR5	Object Tracking	Track workers across frames using Norfair to maintain continuity of detection.	Medium
FR6	Log Non-Compliance Events	Record timestamps, worker IDs, and violations in a database for further analysis.	Medium
FR7	Generate Reports	Create detailed reports of compliance and non-compliance incidents, including timestamps.	High
FR8	Notifications	Send real-time alerts (e.g., email/SMS) to managers when violations occur.	Medium
FR9	System Configuration	Allow administrators to configure detection parameters, add/remove cameras, and update datasets.	Medium
FR10	Dataset Management	Enable uploading and training datasets to improve detection accuracy.	Medium

Non-functional:

ID	Requirement	Description	Priority
NFR1	Performance	The system must process video in real-time with minimal latency (<2 seconds per frame).	High
NFR2	Scalability	The system must handle multiple cameras and large datasets efficiently.	Medium
NFR3	Usability	The system interface must be user-friendly and intuitive for managers and administrators.	Medium
NFR4	Reliability	Ensure high accuracy (>95%) in detecting compliance and non-compliance.	High
NFR5	Security	Protect data (video footage, logs, and reports) with encryption and secure access controls.	High
NFR6	Maintainability	The system must allow easy updates to YOLO, Norfair, and OpenCV modules.	Medium
NFR7	Compatibility	The system must integrate seamlessly with existing CCTV or IP cameras.	Medium
NFR8	Extensibility	The system must allow adding detection for new safety equipment or compliance rules.	Medium
NFR9	Logging	Maintain detailed logs of system activity for troubleshooting and auditing.	Medium
NFR10	Availability	Ensure 24/7 uptime with minimal downtime for maintenance.	Medium

6.2 Architecture overview

The architecture of the **WorkWithHelmtOnly** system is designed to efficiently monitor workplace safety by detecting and evaluating worker compliance with safety standards. This modular and scalable system integrates advanced computer vision technologies and real-time data processing to provide actionable insights into workplace safety compliance.

1. High-Level Components

The architecture is divided into the following key components:

1. **Input Layer**:
    - Captures live video streams or processes uploaded recordings from workplace cameras.
    - Leverages existing CCTV or IP camera systems for easy integration.
2. **Processing Layer**:
    - **Preprocessing Module**:
        - Enhances video frames by resizing, normalizing, and denoising to optimize detection accuracy.
    - **Object Detection Module**:
        - Utilizes YOLO (You Only Look Once) to detect workers and identify helmets and reflective jackets in video frames.
    - **Tracking Module**:
        - Employs Norfair for multi-object tracking, ensuring worker movements are monitored across frames.
3. **Analysis Layer**:
    - **Compliance Evaluation**:
        - Matches detected objects (helmets and vests) against safety standards to identify compliance or violations.
        - Logs violations with timestamps for detailed analysis.
    - **Report Generation**:
        - Produces comprehensive compliance reports, summarizing trends and violations.
4. **Output Layer**:
    - Sends real-time notifications (e.g., email or SMS) for non-compliance events.
    - Stores logs and reports for managerial review and decision-making.

2. System Workflow

1. **Video Input**: The system receives video streams from workplace cameras.
2. **Preprocessing**: Frames are prepared for analysis through enhancements like denoising and normalization.
3. **Object Detection**: YOLO detects workers and safety equipment within video frames.
4. **Tracking**: Norfair ensures workers are consistently tracked across video sequences.
5. **Compliance Evaluation**: The system assesses whether safety protocols are being followed.
6. **Reporting and Notifications**: Non-compliance triggers real-time alerts and generates detailed reports.

6.3 Diagrams

6.3.1 Use Case Diagram

The following Use Case diagram illustrates the interactions between the Work Manager, System Administrator, and Camera System with the workplace safety monitoring system.

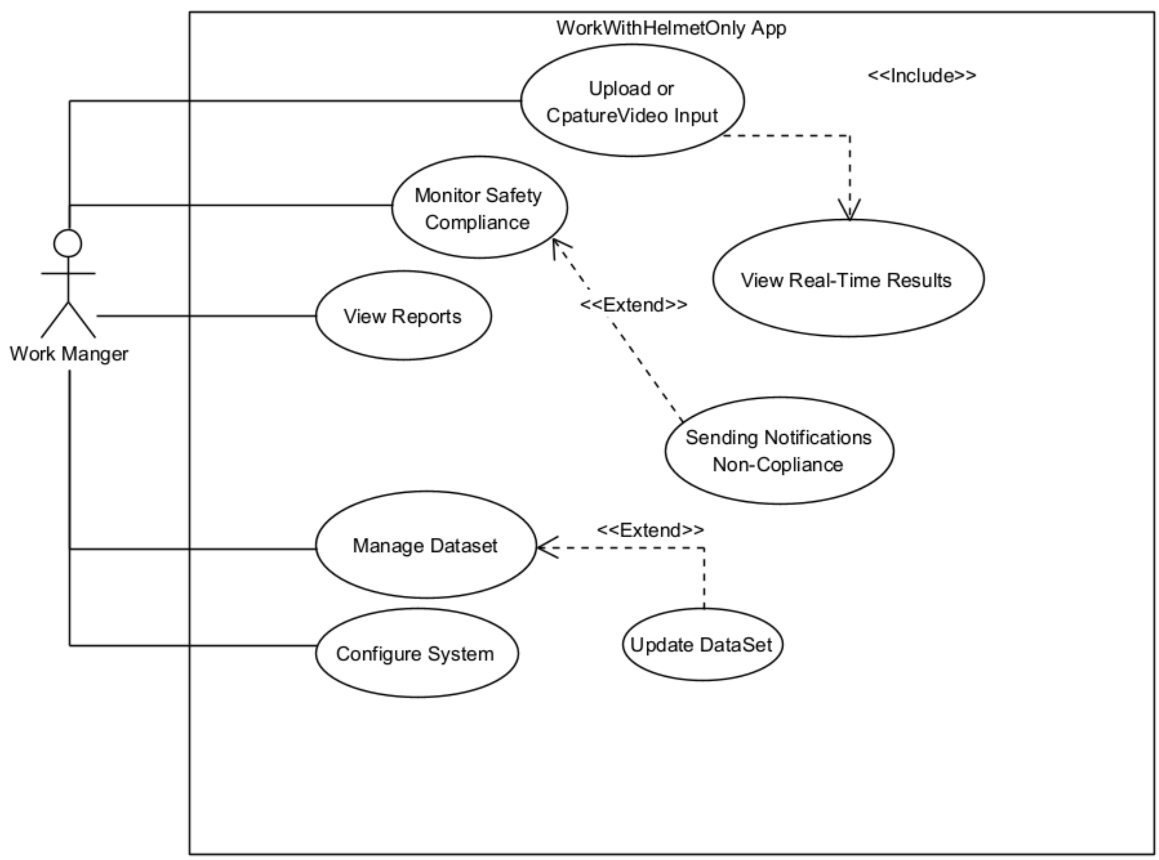

Figure 8. Use Case Diagram

6.3.2 Activity Diagram

This activity diagram represents the flow of processes in the workplace safety monitoring system, which ensures compliance with safety protocols by detecting, tracking, and analyzing workers' use of personal protective equipment (PPE).

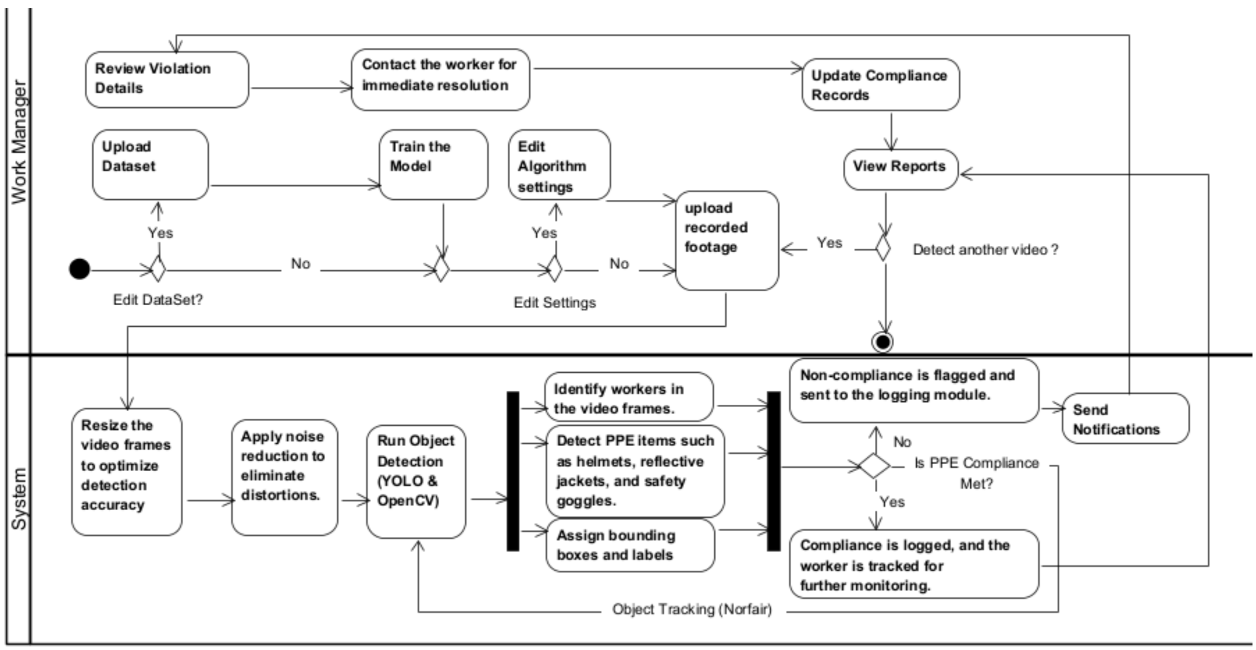

Figure 9. Activity Diagram

6.4 Prototype :

This section showcases the graphical user interface (GUI) designed for the "WorkWithHelmetOnly" application. Each interface is crafted to meet the project's functional requirements, ensuring user-friendly navigation and operational efficiency. The prototype is divided into the following categories:

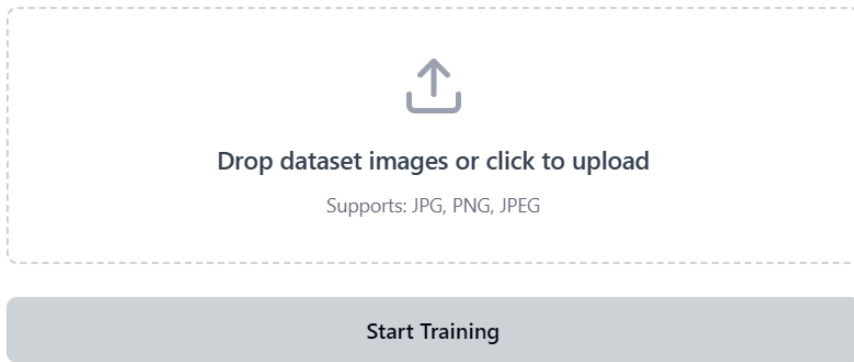

Figure 10. Dataset Upload Interface

Description: The interface displays a file upload area with drag-and-drop functionality for dataset images, alongside a progress bar and training button to initiate model training once the dataset is loaded.

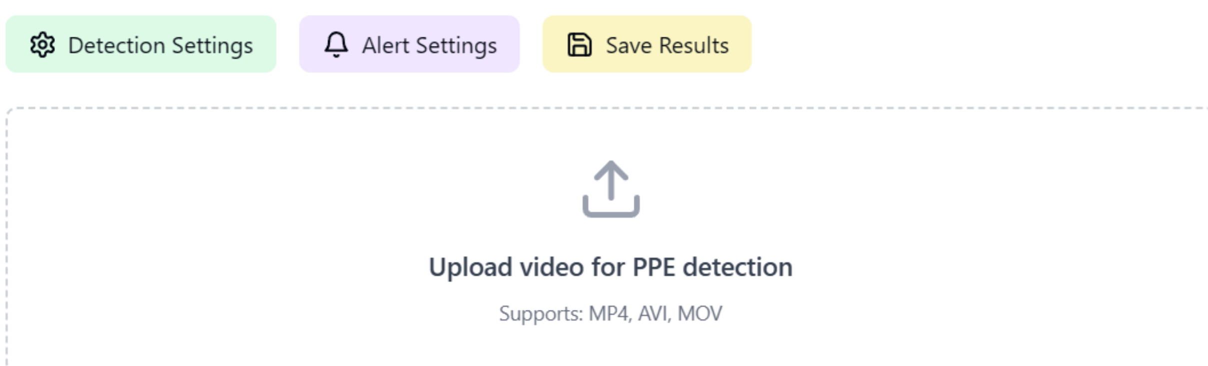

Figure 11. Video Detection Upload Interface

The interface provides video upload functionality with detection controls, settings configuration options, and the ability to export results after PPE violation analysis.

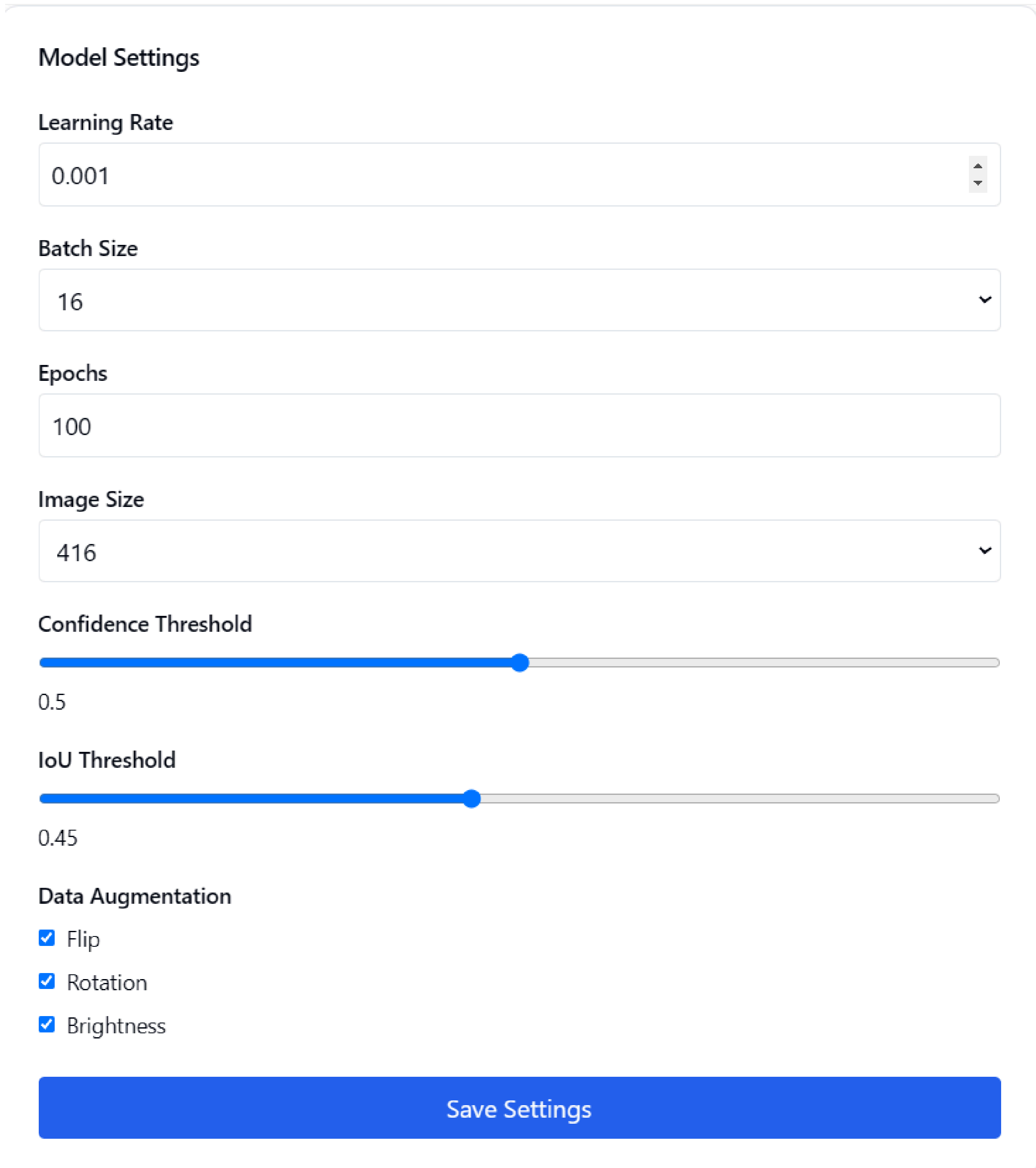

Figure 12. Model Configuration Interface

shows the model configuration interface containing essential parameters like learning rate, batch size, epochs, and data augmentation settings, allowing users to fine-tune the model's training behavior and performance characteristics.

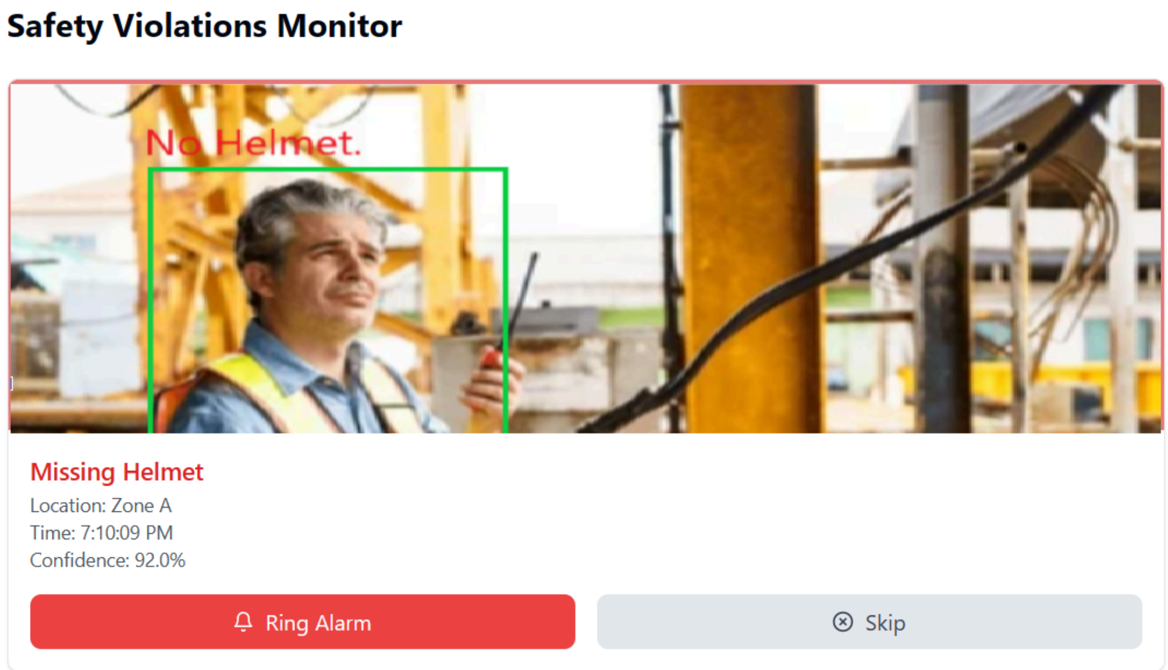

Figure 13: Interface of Reviewing Videos

The interface shows real-time PPE violation detection, highlighting missing safety equipment with bounding boxes and warning labels. Each detection includes location, timestamp, and confidence score, allowing managers to take immediate action through alarm or skip options.

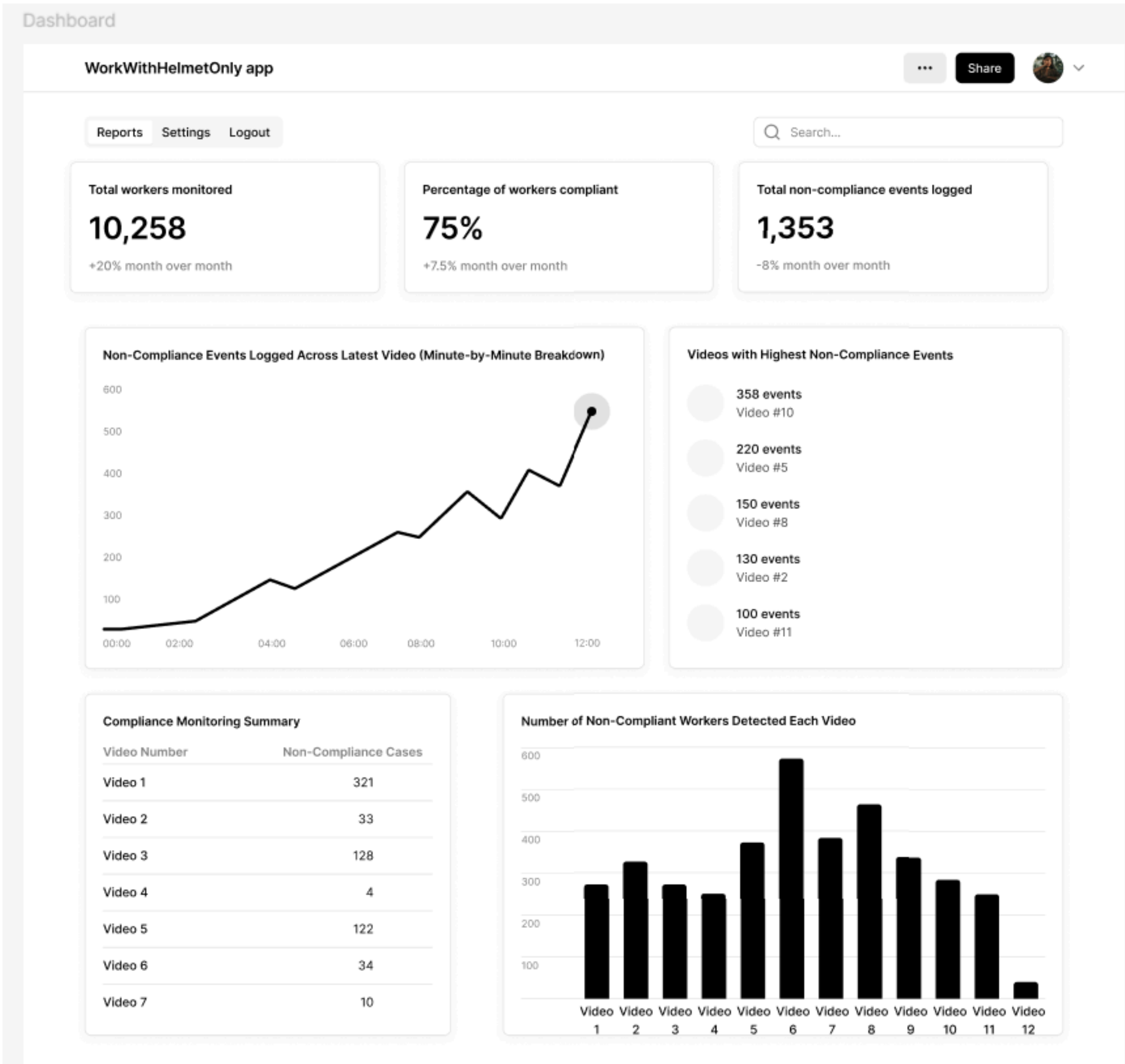

Figure 14. Main Dashboard

The dashboard serves as the central hub for monitoring workplace safety compliance. It displays critical metrics, such as total workers monitored, percentage of compliance, and non-compliance events. The interface also includes visual charts and reports, making it easy for managers to analyze data.

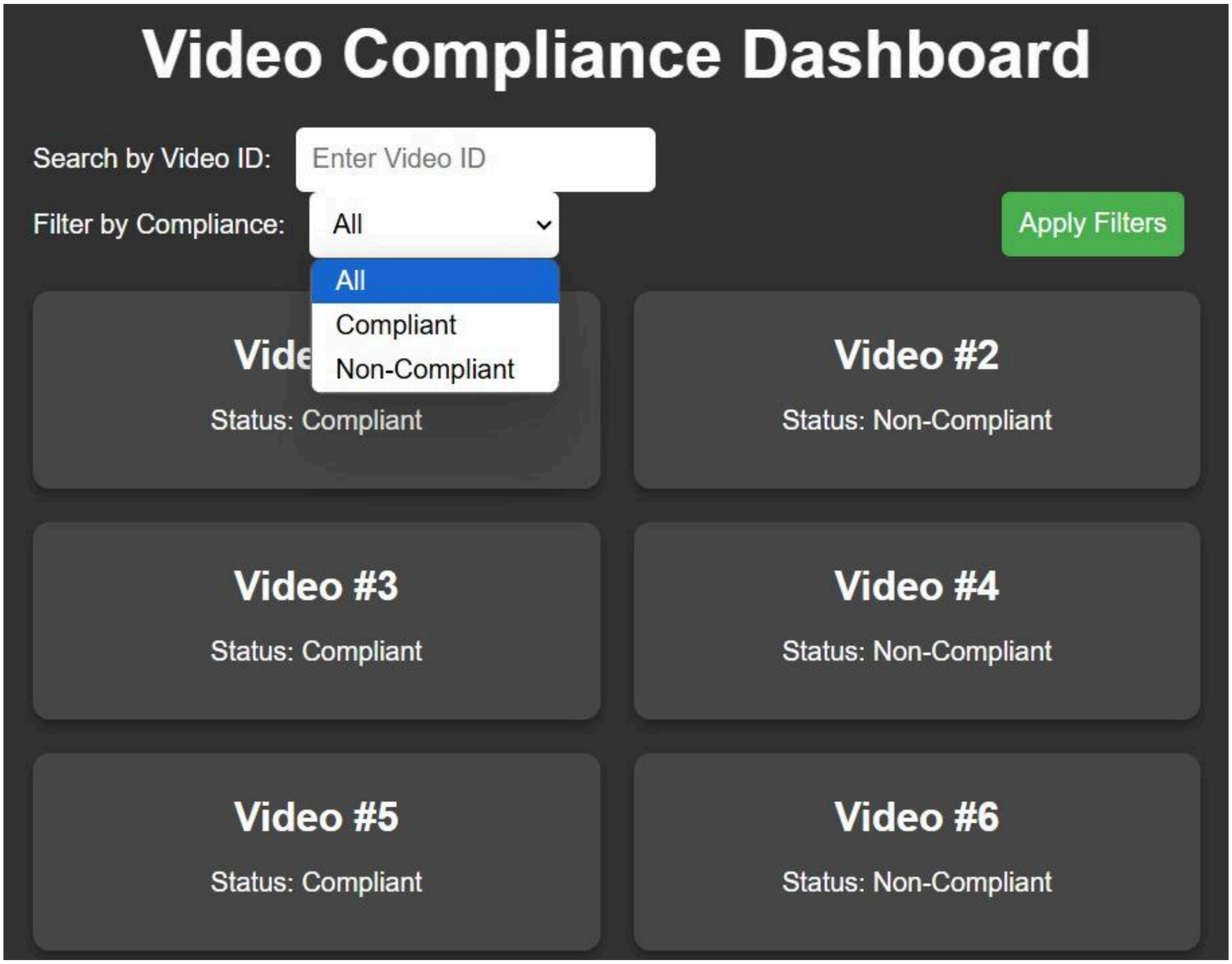

Figure 15. Video Compliance Dashboard

This page allows users to search and filter video compliance statuses by video ID or compliance type. Each video is displayed with its status, ensuring a clear overview of compliance across multiple recordings.

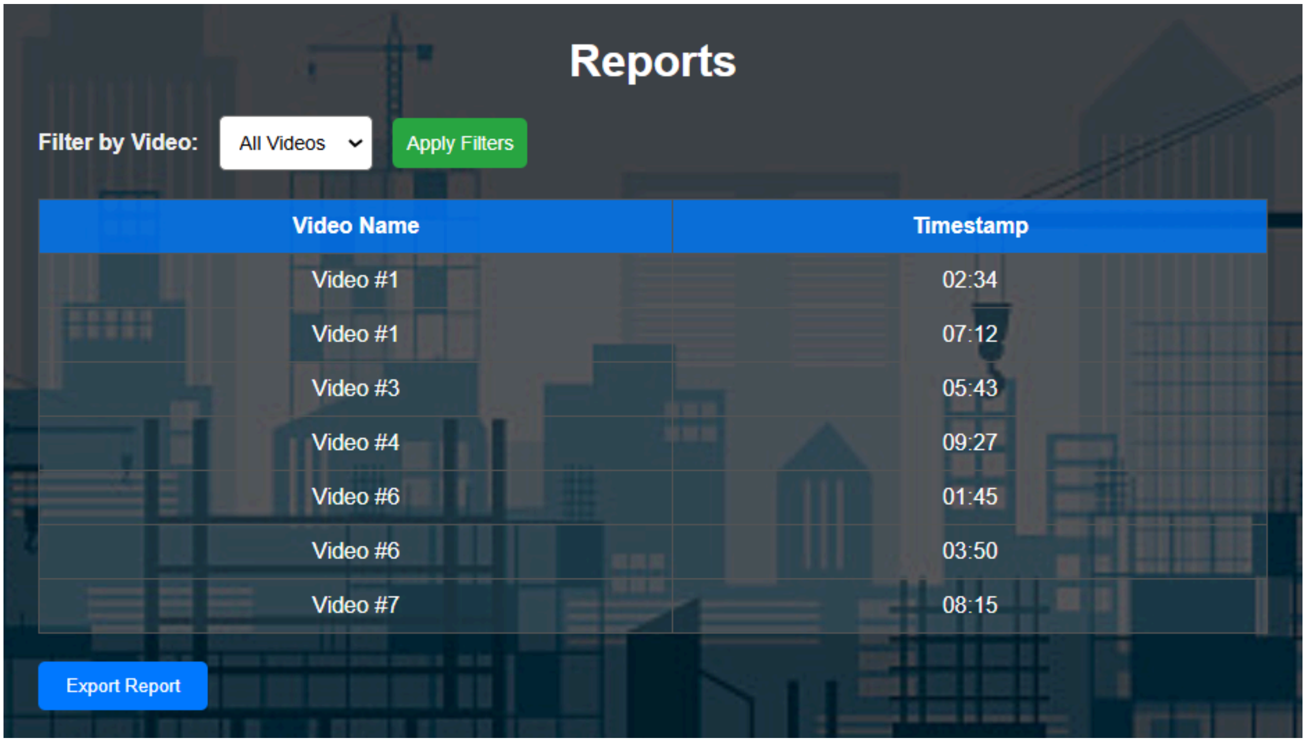

Figure 16. Reports Page

The reports page allows users to view and filter compliance data by video. It lists timestamps and compliance statuses for each video, enabling users to generate detailed reports. The interface includes an export feature for saving reports in external formats.

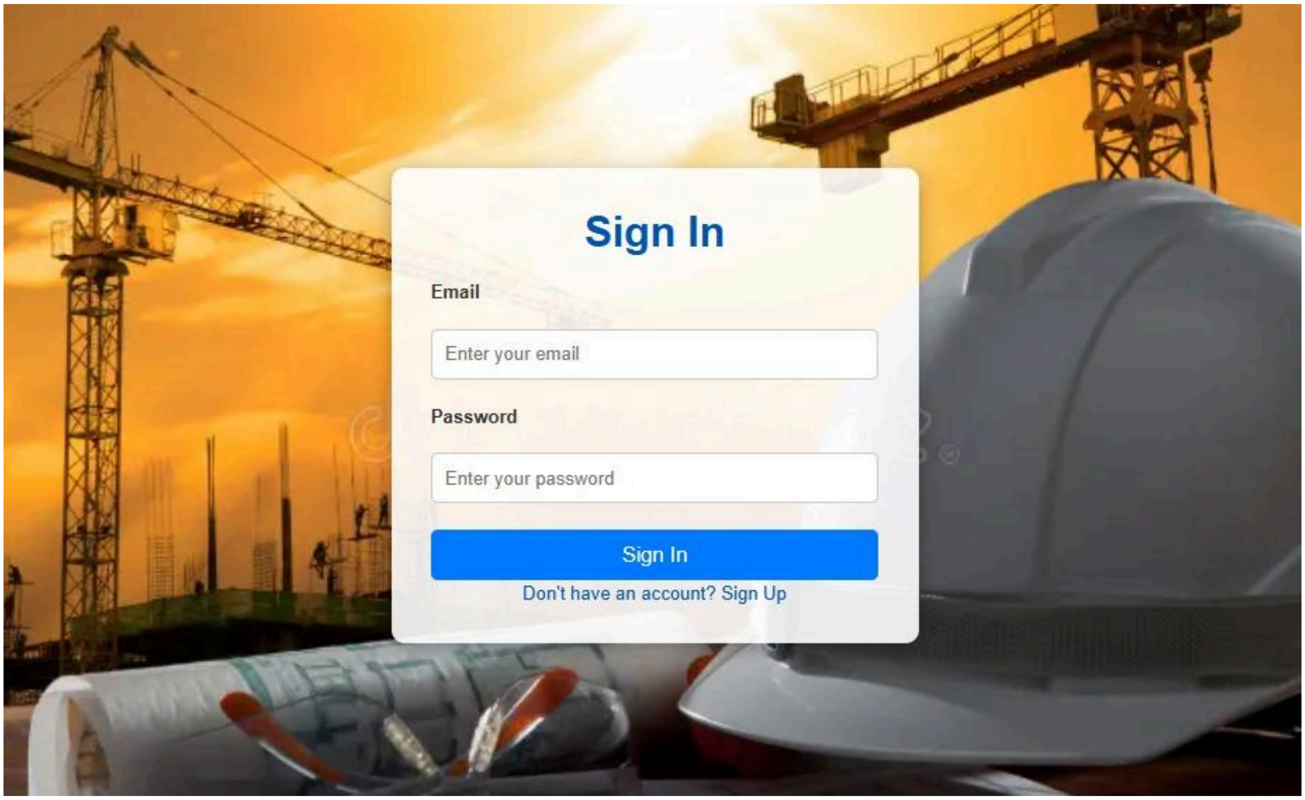

Figure 17. Sign In Page

This page is designed for user authentication. It includes fields for email and password, ensuring secure access to the system. A "Sign Up" option is available for new users, guiding them to the registration page.

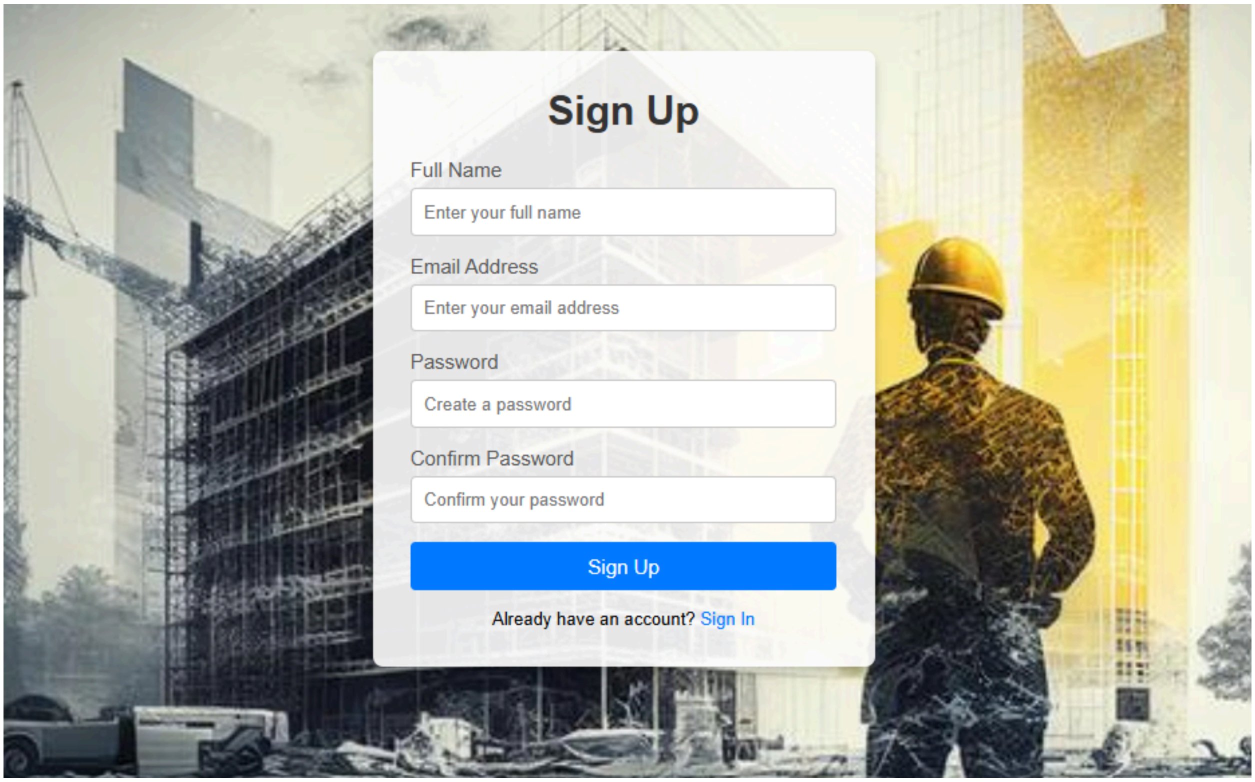

Figure 18. Sign Up Page

The sign-up interface enables new users to create accounts. The form collects essential details, such as name, email, and password. This page ensures easy access for new users while maintaining security protocols.

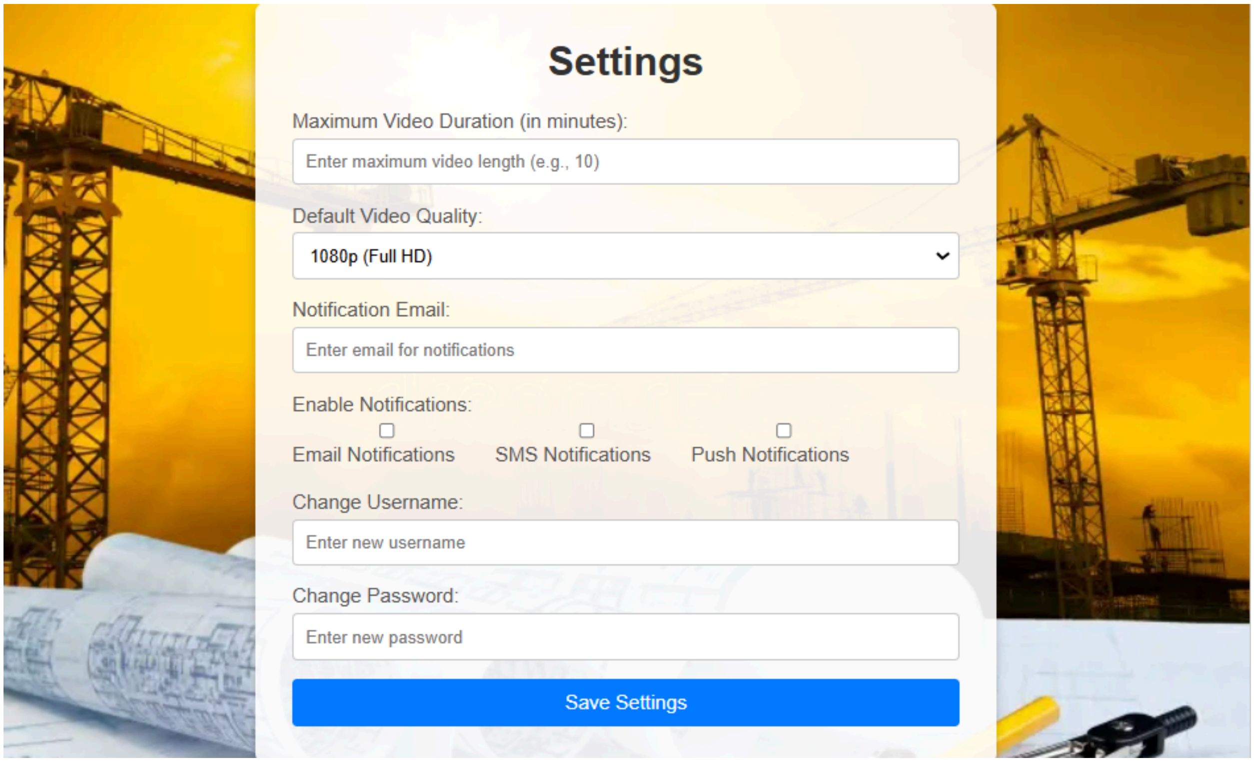

Figure 19. Settings Page

The settings page allows users to configure system parameters, such as video duration, notification preferences, and account settings. This interface ensures flexibility for administrators to tailor the system to their needs.

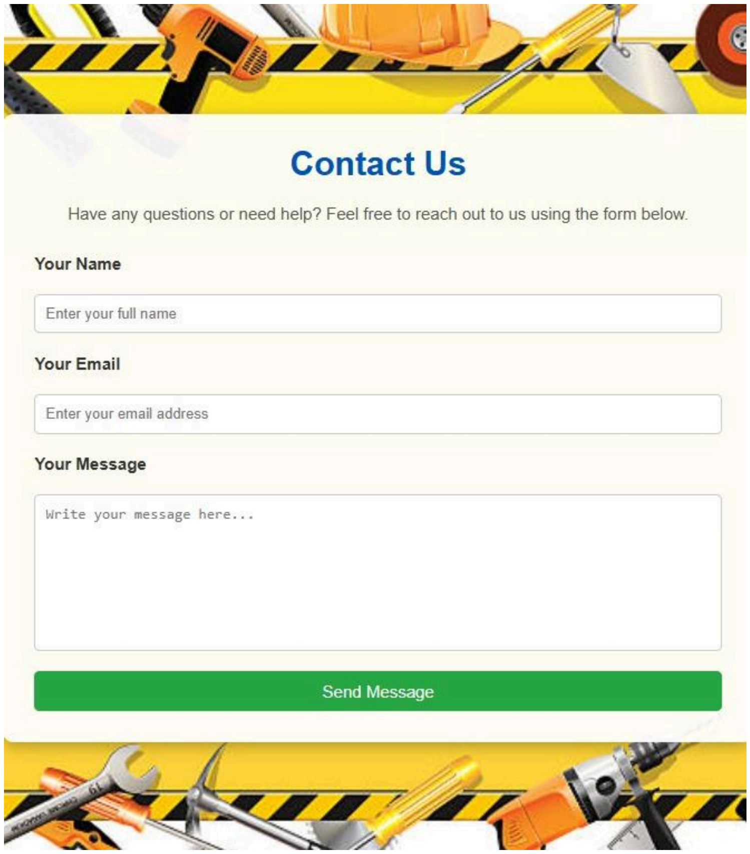

Figure 20. Contact Us Page

This page provides users with a simple form to contact the project team. Users can submit their name, email address, and message. This interface ensures smooth communication between users and the development team, contributing to a better user experience.

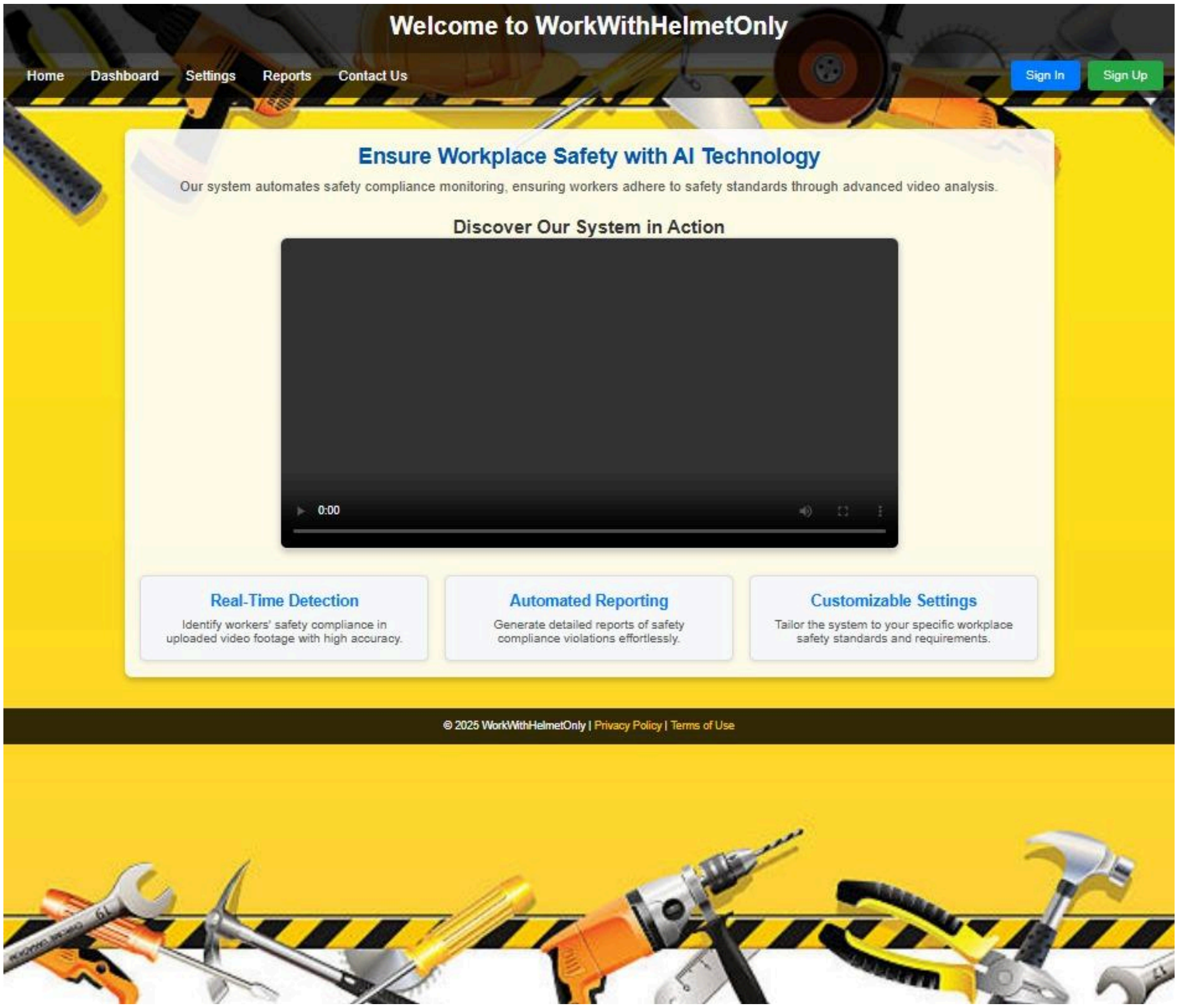

Figure 21. Main Interface of WorkWithHelmetOnly Web

Introduces users to the system with an engaging promotional video and seamless navigation.

7. Verification and Evaluation

7.1 Evaluation

The **WorkWithHelmetOnly** system will be evaluated based on its ability to detect and monitor compliance with safety standards in workplace environments. The system should consistently identify workers, their safety equipment (helmets and reflective vests), and generate detailed compliance reports with high accuracy.

The evaluation criteria will focus on:

- **Detection Accuracy:** The system should achieve over **95%** accuracy in detecting safety gear across various environments.
- **False Positive Reduction:** The system must minimize incorrect flagging of safety compliance.
- **Real-Time Processing:** The system should process video streams in real-time, maintaining a frame rate of at least **30 FPS**.
- **Scalability:** The system must handle multiple video streams simultaneously without performance degradation.
- **Reporting Accuracy:** Reports should correctly reflect all compliance violations and timestamps.

A successful evaluation will be determined by the system's ability to reliably detect missing safety gear under various conditions, including poor lighting and occlusions.

7.2 Verification

The verification phase ensures that all system components perform as expected and meet the project's functional and non-functional requirements. This involves validating each module individually and as part of the integrated system.

Key areas of verification include:

- **YOLO Object Detection:** Accurate identification of helmets and reflective vests.
- **OpenCV Post-Processing:** Proper visualization of compliance and non-compliance.
- **Norfair Tracking:** Consistent tracking of workers across multiple video frames.
- **Data Storage:** Ensuring that compliance data is correctly logged and accessible in the database.
- **Notification System:** Real-time notifications for non-compliance events.

7.3 Testing Plan

In order to evaluate the system performance, the program will be tested with the following scenarios:

Test No.	Test Subject	Expected Result
1	Load an image with workers wearing all required safety gear (e.g., helmet, reflective jacket).	Receive a report indicating full compliance, with no missing items indicated above any worker.
2	Load an image with workers missing a helmet.	Receive a report indicating non-compliance, with 'Helmet Missing' displayed above the corresponding worker's border.
3	Load an image with workers missing a reflective jacket.	Receive a report indicating non-compliance, with 'Reflective Jacket Missing' displayed above the corresponding worker's border.
4	Load an image where no workers are present.	Receive a message stating that no workers were detected in the image.
5	Load an image with multiple workers, some compliant and some non-compliant.	Receive a report indicating compliance for some workers and non-compliance for others, with missing items labeled above the corresponding worker's border.
6	Load an image taken under poor lighting conditions.	The system should process the image and attempt to detect safety tools. An error message appears if the image quality is insufficient.

7	Load an image with workers partially cropped out.	System identifies safety tools for visible portions and provides a partial compliance report with missing items labeled above the corresponding worker's border.
8	Load a video where workers wear safety tools at the beginning but remove them later.	Generate a timeline of compliance changes, with missing items labeled above the worker's border in relevant frames.
9	Test the system with images containing objects that resemble safety tools (e.g., a hat instead of a helmet).	Ensure false positives are minimized; receive a report marking objects that fail compliance criteria with missing items labeled appropriately.
10	Load an image with workers wearing safety tools in unusual colors (e.g., blue helmets).	The system detects safety tools regardless of color, provided they conform to the defined patterns, and labels any missing items above the worker's border.

8. Expected Achievements

This section outlines the key goals and measurable results that the WorkWithHelmetOnly project aims to achieve, emphasizing the technological innovations, practical features, and impact on workplace safety.

8.1 Outcomes

The primary outcome of the WorkWithHelmetOnly project is a robust, automated system capable of detecting and monitoring workers' compliance with personal protective equipment (PPE) standards, specifically helmets and reflective vests. The system will generate detailed compliance reports, including time-stamped evidence of safety violations, and reduce the need for manual safety inspections. This solution aims to contribute to safer work environments, particularly in high-risk industries such as construction and manufacturing.

8.2 Unique Features

Several unique features distinguish WorkWithHelmetOnly from other safety monitoring solutions:

- **Real-Time Object Detection:** Utilizing YOLO for object detection, the system will provide real-time analysis of video streams to detect PPE compliance accurately.
- **Automated Reporting:** The system will automatically generate compliance reports and flag violations without manual input.
- **Scalability:** Designed to work with multiple camera feeds across large industrial spaces.
- **Post-Processing and Tracking:** OpenCV for image enhancement and Norfair for object tracking ensure accurate monitoring even in dynamic environments.
- **Modular Design:** The system's architecture is adaptable, allowing future expansions for additional safety equipment detection and other safety features.

8.3 System Capabilities

The system will be capable of:

- **Real-Time Analysis:** Processing live video feeds and highlighting non-compliance events.
- **Batch Processing:** Analyzing pre-recorded footage for retrospective safety evaluations.
- **Object Detection and Tracking:** Identifying workers and monitoring their compliance with safety standards, including the presence of helmets and reflective vests, across multiple frames.
- **Violation Alerts:** Sending instant notifications via email or SMS when non-compliance is detected.
- **User-Friendly GUI:** The system includes an intuitive graphical user interface (GUI) to enable managers to monitor safety compliance, control workplace situations, generate detailed reports, send alerts, and manage data in the database effectively.

8.4 Impact on Workplace Safety

The system is designed to make a significant impact on workplace safety by:

- **Reducing Injuries:** Preventing accidents through continuous safety monitoring.
- **Minimizing Human Error:** Automating the compliance-checking process to reduce reliance on manual inspections.
- **Encouraging Accountability:** The generated reports and alerts help management address safety concerns promptly.
- **Data-Driven Decisions:** Providing historical data for safety audits and trend analysis, enabling managers to implement data-backed safety policies.

8.5 Compliance with Standards

The project aligns with widely recognized safety guidelines, such as those established by the Occupational Safety and Health Administration (OSHA). It ensures the detection system adheres to best practices in workplace safety monitoring and can be adapted for use in various industries where PPE compliance is critical.

8.7 Success Criteria

The project will be considered successful if it meets the following benchmarks:

- **Detection Accuracy:** Achieves at least 95% accuracy in identifying PPE compliance and non-compliance.
- **Performance:** Real-time analysis with a maximum delay of 2 seconds per frame.
- **Scalability:** The system effectively handles multiple cameras in large workplaces.
- **User Satisfaction:** The system provides clear and actionable reports, while its GUI ensures ease of use, enabling managers to efficiently monitor compliance and reduce safety incidents over time.

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